

VARAD RANE

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EDUCATION

- New Jersey Institute of Technology** Newark, NJ
Master of Science in Computer and Information Science Jan 2025 - May 2026
- Vivekanand Education Society's Institute of Technology** Mumbai, India
Bachelor of Engineering in Electronics; GPA: 3.72 (9.32/10.0) Aug 2018 - May 2022

EXPERIENCE

- Teaching Assistant and Grader** Newark, NJ
New Jersey Institute of Technology Sep 2025 - Dec 2025
 - Grading:** Graded assignments, quizzes, and midterms for CS 116 (C++) and CS 630 (Operating Systems) for 30–40 students via Canvas & Gradescope.
 - Tutoring:** Led weekly 1-hour office hours to clarify concepts and debug assignments and provided timely email support to students for maintaining their learning momentum.
- Deputy Manager - IT Software** Mumbai, India
Kotak Securities Jul 2022 - Aug 2024
 - AD Authentication:** Streamlined Active Directory authentication by introducing token-based sessions, increasing request throughput by 30% and tightening access control across user-facing APIs.
 - Notifications API:** Designed and built an Alert-Management service using .NET/SQL which notified the concerned team instantly when legacy services were overloaded which helped lowering downtime by 15%.
 - API Modernization:** Packaged legacy DB-access logic into a drop-in .NET library by applying SOLID principles and Cache Optimization, removing two network hops, improving latency and resolving socket-exhaustion.
 - Parallel Queue Processing:** Developed Angular + .NET full-stack applications with queue-based processing to aggregate and serve data from multiple SQL databases, improving data retrieval throughput and reliability for high-concurrency workloads.
- Senior Developer** Mumbai, India
Renaissance Cell, VESIT Jan 2021 - Aug 2021
 - Auth & Upload Portal:** Implemented college-email based authentication for secure student verification and a bulk uploader for digitally-signed certificates, ensuring verified issuance and greatly reducing administrative effort.
 - Website & Discord Bot:** Built a website and Discord chat-bot to distribute extracurricular certificates on-demand, improving accessibility and student engagement.

PROJECTS AND PUBLICATIONS

- Privguard Sentinel:** Built a GitHub bot that reviews PRs with an LLM, posting inline comments from structured JSON output; handles large diffs with chunking and configurable severity filtering.
- Stream Chat Organizer:** Designed a chrome extension to add visual elements to YouTube live chat for better readability and improved interaction.
- LazyCurl:** Built a Go-based TUI tool for API stress-testing with curl. Supports throughput/latency benchmarking, JSON handling, session history, and Postman Collections interop.
- Classifying Surface Cracks using Deep CNN:** Published a research paper, combining my knowledge and experience of Deep Learning, Python, and Android App Development

SKILLS

Programming Languages	C#, SQL, Python, JavaScript, TypeScript, GoLang, C, C++, Bash
Frameworks & Libraries	ASP.NET Core, Node.js, AngularJS, SvelteKit, TensorFlow, OpenCV, Pandas, Tailwind CSS, SASS/SCSS
Cloud/DevOps	AWS, Azure, Docker Stack, Kubernetes, Nginx, GitHub Actions
Databases	PostgreSQL, MS-SQL, MySQL, MongoDB, Firebase
Tools	Git, Postman, Figma, Puppeteer, FFmpeg, MATLAB

ACTIVITIES AND LEADERSHIP

- Technical Lead, Executive Board:** Google Developer Student Clubs (GDSC), NJIT.
- Student Tech Educator:** Organized and led multiple workshops on AI, MCP, and VS Code powered by GitHub Copilot and current Microsoft technologies for student audiences.
- Workshop Instructor:** Delivered multiple Robotics lectures with a hands-on lab to grow interest in electronics.